Shahjalal University of Science and Technology
Department of Computer Science and Engineering
3rd Year 2nd Semester Final Examination' November 2019 (Session: 2017-18)
Course Code: CSE 329 Credits: 3 Course Title: Computer Architecture
Time: 3 hrs Total Marks: 100

Group A
[Answer all the questions]

1. Answer any FIVE **5x2= 10**

a) What is the meaning of the flag Z, O and N?
b) Define virtual memory? What is the advantage of using virtual memory?
c) According to the trend in device scaling historically observed by Moore's Law, the number of transistors on a chip in 2025 should be how many times the number in 2015?
d) Draw a block diagram for Von Neumann Architecture.
e) List the five classic components of a computer system.
f) What is stored programming concept?
g) What MIPS instruction does this represent? Choose from one of the four options below.

op	rs	rt	rd	shamnt	funct
0	8	9	10	0	34

h) State Amdahl's Law.

2. Answer any FOUR **4x5= 20**

a) Using a 4-bit version of the algorithm to save pages, let's try dividing 7_{ten} by 2_{ten}, or $0000\ 0111_{two}$ by 0010_{two}. **5**

b) Show the IEEE 754 binary representation of the number -0.75_{ten} in single and double precision. **5**

c) What is the assembly language statement corresponding to this machine instruction? $00af8020_{hex}$ **5**

d) What do you mean by Control hazard? Describe the common methods to overcome control hazard. **5**

e) A compiler designer is trying to decide between two code sequences for a computer. Th e hardware designers have supplied the following facts: **5**

	CPI for each instruction class		
	A	B	C
CPI	1	2	3

For a high-level language statement, the compiler writer is considering two code sequences that require the following instruction counts:

	Instruction counts for each instruction class		
Code Sequence	A	B	C
C-1	2	1	2
C-2	4	1	1

Which code sequence executes the most instructions? Which will be faster?
What is the CPI for each sequence?

f) Describe basic MIPS addressing modes. **5**

3. Answer any TWO **2x10 =20**

a) Construct a 32-bit adder with overflow detection. **10**
b) What do you mean by Datapath? Briefly explain and draw the diagram of the pipelined version of the Datapath. **10**
c) Design a 32-bit ALU that can perform (A AND B), (A OR B), A+B, A-B, A==B, A < B, Overflow Detection, where A and B is 32-bit numbers. **10**

Group B

[Answer all the questions]

4. Answer any FIVE **5x2 =10**

a) What do you mean by ISA? Write some common ISA.
b) Mention the Fields and Field size for R-type, I-type and J-type instructions.
c) What is the difference between Von Neumann architecture and Harvard architecture?
d) What do you mean by Instruction and Instruction set?
e) Negate 2_{ten}, and then check the result by negating -2_{ten}. (use 16 bit)
f) What do you mean by Computer Architecture? What is microarchitecture?
g) Draw the Memory hierarchy.
h) What is underclocking and overclocking? When underclocking is useful?

5. Answer any FOUR **4x5 =20**

a) Using 4-bit numbers to save space, multiply $2_{ten} * 3_{ten}$, or $0010_{two} * 0011_{two}$. **5**

b) What do you mean by memory? Implement a 1-bit gated latch using this latch design an 8-bit register. **5**

c) Briefly describe the steps that transform a program written in a high-level language such as C into a representation that is directly executed by a computer processor. **5**

d) Assume that i and k correspond to registers $s3 and $s5 and the base of the array save is in $s6. What is the MIPS assembly code corresponding to this C segment? **5**

```
while(save[i] == k) {
        i += 1
}
```

e) A program runs in 10 seconds on computer A, which has a 2 GHz clock. We are trying to help a computer designer build a computer, B, which will run this program in 6 seconds. Th e designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing computer B to require 1.2 times as many clock cycles as computer A for this program. What clock rate should we tell the designer to target? **5**

f) If a company has 20,000 computers, each with a MTTF of 20 days and it experiences catastrophic failure only if 1/5 of the computers fail, what is the MTTF for the system? **5**

6. Answer any TWO **2x10 =20**

a) i. What do you mean by cache miss? Draw the block diagrams of an eight-block cache configured as direct mapped, two-way set associative, four-way set associative, and fully associative. **5+5**

ii. Suppose a program takes 1 billion instructions to execute on a processor running at 5GHz. Suppose also that 40% of the instructions execute in 3 clock cycles, 50% execute in 4 clock cycles and 10% execute in 5 clock cycles. What is the execution time for the program or task?

b) What is data hazard? Explain with an example. Describe the following techniques with proper diagrams. **10**

- Pipeline Stall
- Reordering Code
- Forwarding

c) Let's tackle a recursive procedure that calculates factorial: **10**

```
int fact (int n)
{
    if (n < 1) return (1);
    else return (n * fact(n - 1));
}
```

What is the MIPS assembly code?